

SOLUTION

1. (a)

$$\lambda_{\text{obs}} = \lambda_{\text{source}} \sqrt{\frac{1 - \beta}{1 + \beta}} \quad (1)$$

$$\beta = \frac{1 - \left(\frac{\lambda_{\text{obs}}}{\lambda_{\text{source}}}\right)^2}{1 + \left(\frac{\lambda_{\text{obs}}}{\lambda_{\text{source}}}\right)^2} \quad (2)$$

$$\frac{\lambda_{\text{obs}}}{\lambda_{\text{source}}} = \frac{4}{5} \quad (3)$$

$$\beta = \frac{9}{41} \quad (4)$$

$$(5)$$

The speed with which you must approach the red light for it to appear green is $\frac{9}{41}c$.

First formula, [1 pt], final ans. [1 pt], for cal. mistake [0.5 pt]

(b) Using

$$\lambda_{\text{obs}} = \lambda_{\text{source}} \sqrt{\frac{1 - \beta}{1 + \beta}} \quad (6)$$

$$\lambda_{\text{obs}} = 570 \times \sqrt{\frac{1 - 9/41}{1 + 9/41}} \quad (7)$$

$$\lambda_{\text{obs}} = 456 \text{ nm} \quad (8)$$

The yellow light would appear blue.

Identify λ_s : [1 pt] and caln [1 pt]

2. (a) The chain rule gives us

$$dx = \frac{\partial x}{\partial t} dt + \frac{\partial x}{\partial \phi} d\phi = 2a \cos \phi dt - 2at \sin \phi d\phi \quad (9)$$

$$dy = \frac{\partial y}{\partial t} dt + \frac{\partial y}{\partial \phi} d\phi = 2a \sin \phi dt + 2at \cos \phi d\phi \quad (10)$$

$$dz = \frac{\partial z}{\partial t} dt + \frac{\partial z}{\partial \phi} d\phi = 2at dt \quad (11)$$

$$ds^2 = dx^2 + dy^2 + dz^2 \quad (12)$$

$$= (2a \cos \phi dt - 2at \sin \phi d\phi)^2 + (2a \sin \phi dt + 2at \cos \phi d\phi)^2 + (2at dt)^2 \quad (13)$$

$$= 4a^2(1 + t^2)dt^2 + 4a^2t^2d\phi^2 \quad (14)$$

Chain rule: [0.5 pt], Doint it correctly [1 pt] and rest cal [1 pt]

(b)

$$\Gamma_{tt}^t = \frac{t}{1+t^2}, \quad \Gamma_{\phi\phi}^t = -\frac{t}{1+t^2}, \quad \Gamma_{t\phi}^\phi = \frac{1}{t} \quad (15)$$

Hence the geodesic equations can be written as

$$\ddot{t} + \frac{t}{1+t^2}\dot{t}^2 - \frac{t}{1+t^2}\dot{\phi}^2 = 0 \quad (16)$$

$$\ddot{\phi} + 2\frac{1}{t}\dot{\phi}\dot{t} = 0 \quad (17)$$

Each equation [1 pt.] If done only gammas correctly then [1 pt]. If gammas wrong and equation wrong but correct procedure [0.5 pt.]

3.

$$g_{,n}^{im} = (g^{mk} g^{il} g_{kl})_{,n} \quad (18)$$

$$g_{,n}^{im} = \delta_k^i g_{,n}^{mk} + \delta_l^m g_{,n}^{il} + g^{mk} g^{il} g_{kl,n} \quad (19)$$

$$g_{,n}^{im} = 2g_{,n}^{im} + g^{mk} g^{il} g_{kl,n} \quad (20)$$

Simplifying the above equation, we get the desired equation.

OR

$$g^{im} g_{mk} = \delta_k^i \quad (21)$$

$$(g^{im} g_{mk})_{,n} = 0 \quad (22)$$

$$g_{,n}^{im} g_{mk} + g^{im} g_{mk,n} = 0 \quad (23)$$

$$g_{,n}^{im} = -g^{mk} g^{im} g_{mk,n} \quad (24)$$

The first equation is the definition of the inverse metric tensor. The second eqn is obtained by differentiating. The third eqn is just the chain rule. The fourth eqn is by multiplying with the inverse metric tensor. **[1 pt] for steps in case not got the correct answer. Incomplete answers [0.5 pt].**

4. (a)

$$\Gamma_{\theta\phi}^{\theta} = \Gamma_{\phi\theta}^{\theta} = -\frac{r \sin \phi}{(R + r \cos \phi)} \quad (25)$$

$$(26)$$

$$\Gamma_{\theta\theta}^{\phi} = \frac{(R + r \cos \phi) \sin \phi}{r} \quad (27)$$

[1pt] for each gammas.

- (b) The only non zero components are $R_{\theta\phi\theta}^{\phi}$ and $R_{\phi\phi\theta}^{\theta}$. Substituing the above chritoffel symbols into the expression for the Riemann tensor we get

$$R_{\theta\phi\theta}^{\phi} = \Gamma_{\theta\theta,\phi}^{\phi} - 0 + 0 - \Gamma_{\theta\lambda}^{\phi} \Gamma_{\theta\phi}^{\lambda} \quad (28)$$

$$= \Gamma_{\theta\theta,\phi}^{\phi} - \Gamma_{\theta\theta}^{\phi} \Gamma_{\theta\phi}^{\theta} \quad (29)$$

$$= \frac{(R + r \cos(\phi)) \cos(\phi)}{r} \quad (30)$$

$$R_{\phi\phi\theta}^{\theta} = \Gamma_{\theta\theta,\phi}^{\theta} - 0 + \Gamma_{\phi\lambda}^{\theta} \Gamma_{\theta\phi}^{\lambda} - 0 \quad (31)$$

$$= \Gamma_{\theta\theta,\phi}^{\theta} + \Gamma_{\phi\theta}^{\theta} \Gamma_{\theta\phi}^{\theta} \quad (32)$$

$$= -\frac{r \cos(\phi)}{R + r \cos(\phi)} \quad (33)$$

The other components can be argued to be 0 as follows: $R^i jkk = 0$ due to the antisymmetry in the last 2 indices

$$R_{\theta jk}^{\theta} = g^{\theta\theta} R_{\theta\theta jk} + g^{\theta\phi} R_{\theta\phi jk} \quad (34)$$

$$= 0 \quad (35)$$

Here $R_{\theta\theta jk} = 0$ because of the antisymmetry in the first 2 indices and the metric is diagonal leading to both the terms vanishing. The same case occurs for $R_{\phi jk}^{\phi}$. OR

Alternatively, just counting the number of components for R_{ijkl} are $N^2(N^2 - 1)/12$ gives one component which is $R_{\theta\phi\theta\phi}$. Using this and the symmetry properties, one can get $R_{\phi\phi\theta\theta}^{\theta}$ and $R_{\theta\theta\phi\phi}^{\phi}$.

Argue from counting formula that for 2-D, only one R_{ijkl} is non-zero [1 pt]. Compute the value [1pt]. Attempt to compute with correct expression [1pt].